

HSIC Versus USB

Andrew Rogers
is applications
engineer at
Microchip Technology

High-speed inter-chip (HSIC) interface is becoming more popular due to its notable advantages over USB for hard-wired inter-chip applications. The interface is a two-signal, source-synchronous interface that can provide USB high-speed data at 480Mbps. Data transfers are 100 per cent host-driver compatible with traditional USB topologies. Full-speed (FS) and low-speed (LS) are not supported by the format. However, a hub with HSIC can provide FS and LS support.

The interface differs from USB in the physical layer only. Significant features include no chirp protocol, source-synchronous serial data transmission and no hot removal or attach as the interface is always connected.

It has 1.2V signal levels designed for low-power applications at standard LV

CMOS levels. Maximum trace length is 10cm. The protocol for data transactions between host and device via HSIC is the same as USB, as shown in Fig. 1.

The primary difference between the two is that in HSIC all information is transmitted via a single data line, and a strobe signal communicates when to sample the received data signal. HSIC uses double data rate (DDR) signalling; data are sampled at both the rising and falling edges of the strobe signal. The strobe signal oscillates at a frequency of 240MHz, which provides a total data rate of 480Mbps.

Advantages of HSIC over USB

HSIC has significant advantages over USB. For a start, it is a fully-digital standard and, thus, no analogue frontend is required. Lack of an analogue frontend means die sizes can be reduced and, thus, so can the cost. Additional die reduction can also be made due to the decreased amount of digital logic required by the simplified connection protocol.

HSIC standard does not inherently reduce power consumption, but removal of the analogue frontend can lead to lower-power designs, especially since analogue circuitry does not necessarily scale one-to-one with digital circuits for reductions in process feature size.

HSIC is especially low-power when placed into the suspended state as there is no current drawn on the strobe or data lines. By comparison, standard USB draws a minimum of 200µA on D+ through a 1.5kΩ pull-up resistor when suspended.

Because HSIC is only different from USB at the physical layer, migrating from USB to HSIC is not like changing to a completely new standard. This means existing USB software stacks and USB protocol knowledge bases can be quickly transitioned to HSIC.

Data sampling

With standard USB, every data packet begins with a sync pattern to allow

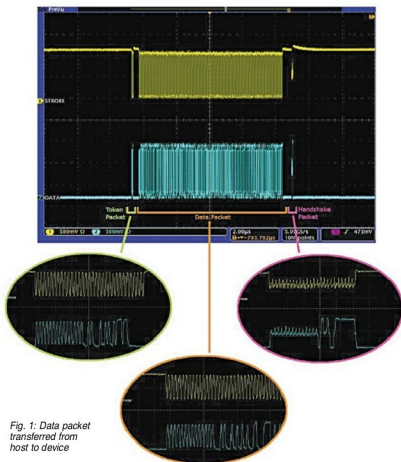


Fig. 1: Data packet transferred from host to device